

CAPSTONE PROJECT PROPOSAL

Predictive Maintenance for KPC Pipeline Pump Infrastructure

1. Team Members

Name	Role
BRIAN KIOKO	Data Engineering Lead : pipeline, synthetic data generation, ETL
SILAS KIBET	Statistical & ML Lead : hypothesis testing, feature engineering, modeling
INGRID MIRIAM	Dashboard & Visualisation Lead : Dashboarding, stakeholder views
EUGENE OBUNDE	Storytelling & ROI Lead : briefs, executive narrative, business case
LAMECK MUGO	QA & Documentation Lead : testing, UAT notes, deployment readiness

2. Problem Statement

In Kenya's refined-petroleum pipeline network (Kenya Pipeline Company, 1,342 km spanning Mombasa–Nairobi–Nakuru–Eldoret–Kisumu), the problem of unplanned pumping-unit failures at booster and depot stations leads to throughput stoppages, deferred fuel deliveries to OMCs, emergency repair costs, and safety-critical pressure excursions. Currently, time-based, fixed-interval maintenance is used, which fails because it ignores each pump's actual degradation state — servicing healthy machines wastefully while missing machines that are genuinely about to fail. We propose using statistical hypothesis testing, rolling-window sensor feature engineering, and classification/regression modeling (failure-risk prediction and Remaining Useful Life estimation) to flag at-risk pumps days in advance and shift the maintenance workflow from calendar-based to condition-based scheduling, aiming to achieve an 80%+ failure-catch rate, under 5% false alarms, and a measurable reduction in unplanned downtime hours fleet-wide.

3. Domain & Context

Industry: Oil & Gas — midstream pipeline operations and predictive maintenance.

Scenario: KPC operates approximately 1,342 km of pipeline with a combined storage/handling capacity of roughly 14 billion litres annually, moving refined product through 10 pumping stations from Mombasa to the Nairobi terminal and onward to Nakuru, Eldoret, and Kisumu. KPC has invested in Variable Frequency Drive (VFD) pump technology, which generates the flow,

pressure, temperature, motor current, and vibration telemetry this project models. Pipeline safety incidents in Kenya — including major fires linked to infrastructure and valve failures, and documented product spills such as an estimated 400,000 litres at Thange River (2015) and 551,000 litres at Kiboko (2018, valued at approximately KES 63 million) — demonstrate that undetected mechanical and pipeline degradation is a documented safety and financial risk, not an abstract concern.

This capstone builds directly on the team's Hackathon #1 project (order-to-cash billing reconciliation, Domain D) but deliberately shifts to a distinct analytical competency — statistical prediction and condition-based maintenance — while remaining in a domain the team already understands operationally through the KPC partnership.

4. Data Source

KPC's real internal pump sensor and maintenance logs are not publicly available. Data strategy is therefore three-layered and fully realistic about this constraint:

- Primary — physics-informed synthetic SCADA data. Generated in-house, parameterised on KPC's public infrastructure figures and realistic pump physics. Validated in Week 6 lab achieving 94.2% accuracy.
- Validation benchmark — NASA C-MAPSS. Canonical public benchmark for Remaining-Useful-Life modeling used to report performance against citable standards.
- Secondary — AI4I 2020 Predictive Maintenance Dataset (UCI). Industrial dataset with 10,000 records used for sanity checks on feature-engineering before scaling.

5. Success Metrics

- Achieve a minimum 80% recall for near-failure detection within a 7-day intervention window (Sensitivity).
- Current Week 6 baseline: 84.3%; the objective is to sustain or exceed this performance on the expanded fleet.
- Restrict false alarm rates to under 5% to uphold technician trust in automated alerts (Baseline: 2.4%).
- Minimize Remaining Useful Life (RUL) prediction error to under 90 hours MAE (improving from 105.9 hours).
- Realized through sophisticated sensor feature engineering and the expansion of the physics-informed dataset.
- Forecast mechanical failure risk at least 3 days prior to the predicted terminal event.
- Providing maintenance personnel with a viable scheduling window for proactive, condition-based intervention.
- Produce audience-calibrated deliverables for each iteration, including technical engineering briefs.
- And CFO-facing ROI narratives, adhering to the communication standards established during the pilot phase.
- Deploy a functional dashboard by Week 7 and a CI/CD-integrated service for the graduation showcase.
- Meeting all technical requirements for operational monitoring and deployment readiness by Week 12.

6. Initial Timeline (Weeks 7–12)

Week	Focus	Deliverable
7	Safety, Compliance & Environmental Analytics	Expand synthetic dataset; build first interactive dashboard
8	Supply Chain & Logistics Analytics	Integrate depot context; refine stakeholder narrative
9	Machine Learning for Operations	Compare LR against ensembles; cross-validate models
10	AI-Assisted Operational Tools	Accelerate iteration via AI tools; stress-test edge cases
11	Analytics Strategy & ROI	Quantify ROI; package technical portfolio
12	Showcase & Graduation	Final demo, stakeholder Q&A, and ROI pitch